

Updates on Project

Daniel Krutz <dxkvse@rit.edu>
To: "Piter Garcia (RIT Student)" <pzg8794@rit.edu>
Cc: "Devroop Kar (RIT Student)" <dk7405@rit.edu>

Tue, Nov 18, 2025 at 3:09 PM

Hi guys, one thing that would be good for you to discuss as well is how you feel about the application of what we're doing and its potential application/building for the quantum domain.

Essentially, it's this idea:
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Quantum entanglement path selection is the process of dynamically identifying and optimizing routes through intermediate quantum nodes to establish high-fidelity entanglement between distant qubits in a quantum network. It is crucial that quantum networks make high-quality, adaptive routing decisions in order to establish entanglement across dynamic, noisy and resource-limited topological [Cite1]. Unfortunately, current state-of-the-art techniques are limited by their inability to be explainable, scalable and policy transferable [Cite2]. This is especially the case trying to balance competing objectives such as maximizing fidelity, minimizing purification overhead, and maintaining coherence lifetimes. The objective of this work is to address the problem of how to evolve interpretable, adaptive routing strategies that can efficiently learn and generalize across varying quantum network conditions.

We will address the issue of current techniques being unable to be explainable, scalable and policy transferable through the adaptation and modification of a hybrid evolutionary machine learning method that combines graph-based genetic programming (GP) with neuroevolution , specifically the EXA-GP framework. This model will provide the benefits of: Adaptive optimization, Explainability, Scalability, Tradeoff management, Robustness under uncertainty.

Thanks
-Dan

On Nov 18, 2025, at 1:31 PM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Yes, that works for me.

On Tue, Nov 18, 2025 at 12:43 PM Devroop Kar (RIT Student) <dk7405@rit.edu> wrote:

Hi,

We can have a call tomorrow around 12:30 if that works for you.

Thanks and regards,
Devroop Kar

On Mon, Nov 17, 2025 at 4:35 PM Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Devroop,

My apologies, I completely forgot to email you with so many deadlines I was keeping track of.

What days work best for you this week? My schedule is flexible since I am currently working on papers, so I can easily work around your schedule.

Best,
Piter

On Mon, Nov 17, 2025 at 2:08 PM Devroop Kar (RIT Student) <dk7405@rit.edu> wrote:

Hello Piter,

Let me know if you have some time this week to go over your progress and results.

Thanks and regards,
Devroop Kar